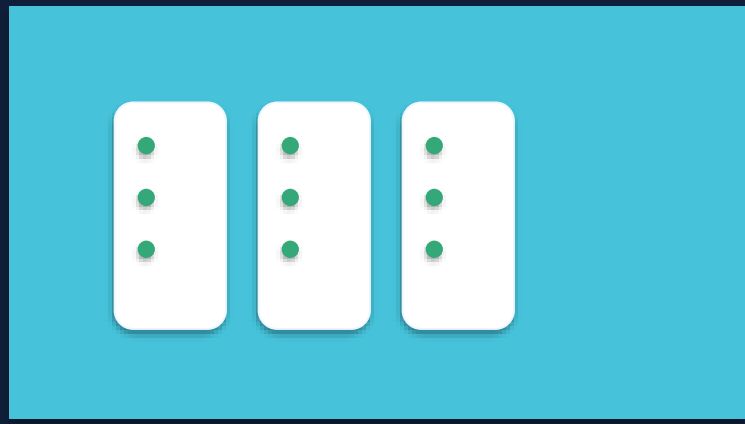
Cloud Migration Strategies

Moving Enterprise Applications to the Cloud



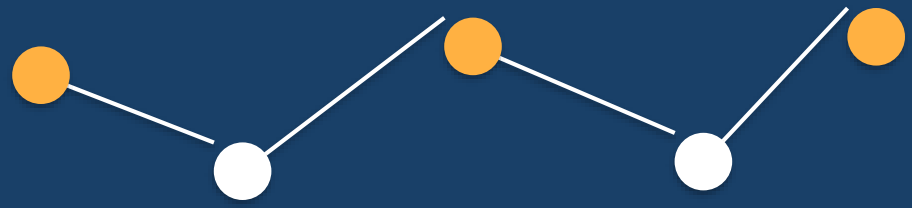
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1. Migration Overview & Strategies

What is cloud migration?

Moving applications, data and workloads from local data centres to cloud infrastructure, with planned testing and cutover.

Common migration approaches

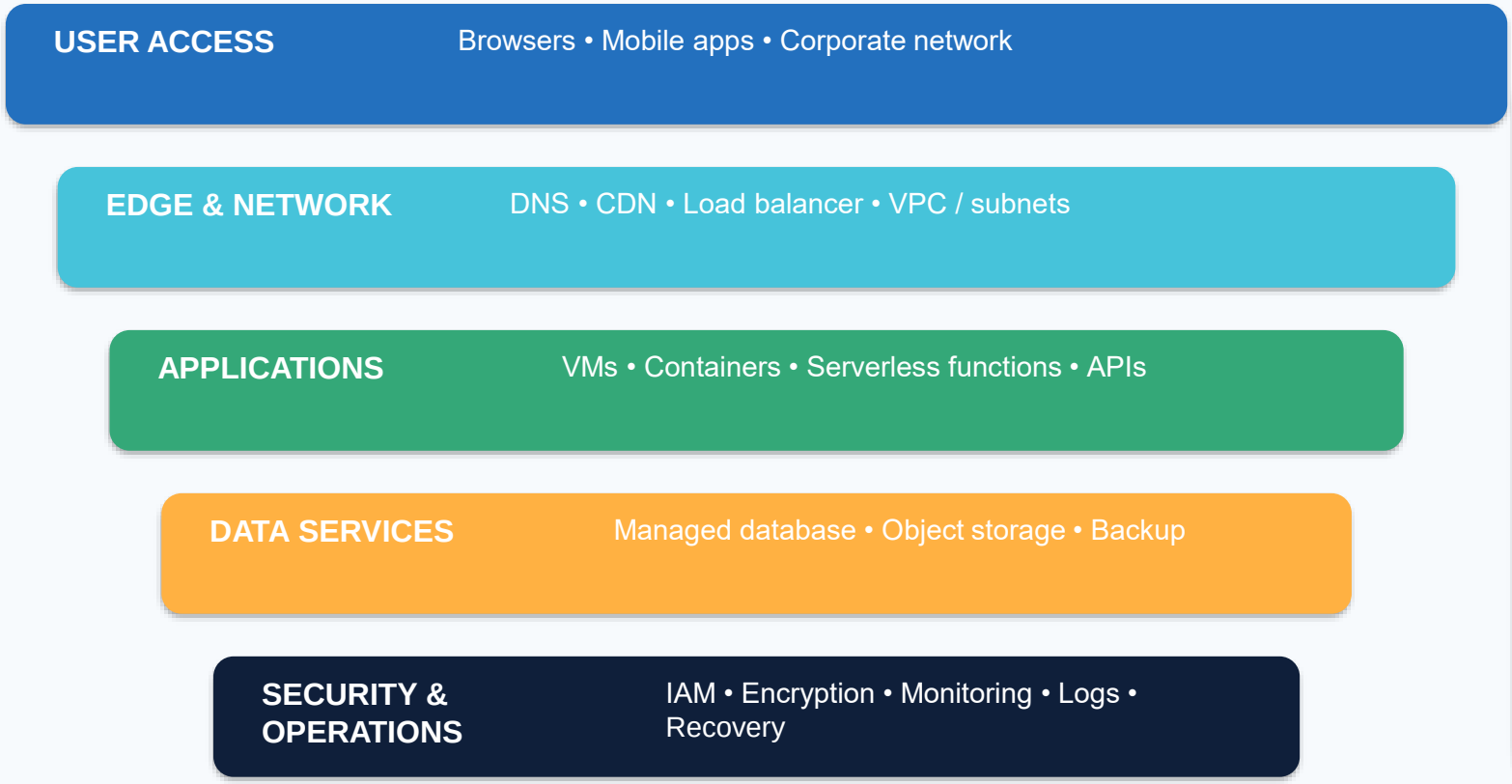
- Rehost (lift-and-shift): move with minimal changes.
- Replatform: make limited cloud optimisations.
- Refactor / re-architect: redesign for cloud-native services.
- Repurchase: replace with a SaaS product.
- Retain or retire: keep for now, or decommission.

Choose strategy by business need

1. **Assess** — Inventory apps, dependencies and risks
2. **Plan** — Choose target and migration wave
3. **Pilot** — Move a low-risk application first
4. **Migrate** — Transfer data, test and cut over
5. **Optimise** — Monitor cost, speed and reliability

2. Cloud Architecture Considerations

Design the target environment before moving production workloads.



Key design checks

- Security: least-privilege access and encryption.
- Availability: multi-zone deployment and backups.
- Performance: latency, capacity and scaling.
- Integration: APIs, identity and legacy systems.
- Governance: compliance, ownership and budgets.

3. Benefits, Challenges & Mitigation

BENEFITS

- **Elastic scale**
  Add or reduce resources with demand.
- **Agility**
  Provision environments faster.
- **Resilience**
  Use redundancy, backups and recovery.
- **Global reach**
  Deploy closer to customers.
- **Managed services**
  Reduce routine infrastructure work.

CHALLENGES + RESPONSES

- **Unexpected costs**
  Budgets, tags, alerts and rightsizing.
- **Security & compliance**
  IAM, encryption, audits and policies.
- **Downtime / data loss**
  Pilot, replicate data and rehearse rollback.
- **Legacy dependencies**
  Map interfaces; migrate in small waves.
- **Skills gap**
  Train teams and document operations.

4. Case Study: Netflix’s Cloud Migration

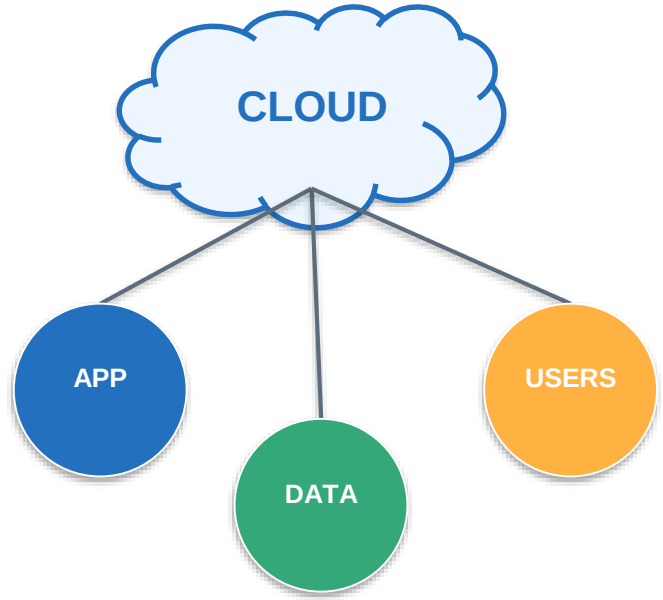
From data centres to cloud-native streaming

Netflix began its cloud journey after a major database failure in 2008. It completed its streaming-service migration in January 2016, using AWS.

Migration journey

- **2008** — Reliability problem prompts cloud journey
- **Next years** — Move systems in stages; build distributed services
- **2016** — Streaming-service migration completed

What made it successful?



Microservices • distributed data • automation

Key lesson: migration was a multi-year transformation—not a simple server copy. Resilience and independent scaling were central.

Source: Netflix, “Completing the Netflix Cloud Migration” (2016).

5. Applications & Conclusion

Where cloud migration is used

Enterprise systems

ERP, CRM, HR and collaboration tools

E-commerce

Product catalogues, checkout and seasonal traffic

Education & research

Learning platforms, labs and data processing

Healthcare & finance

Secure records, analytics and disaster recovery

Success checklist

✓ Clear business goals and application inventory
✓ Secure landing zone, pilot test and recovery plan
✓ Monitoring, cost controls and trained teams

Conclusion

- Cloud migration can improve agility, scalability and resilience.
- There is no single best strategy: match the approach to each application.
- Plan security, costs, data, testing and rollback from the start.
- Migrate in waves, measure outcomes, and continuously optimise.


Thank
you!